

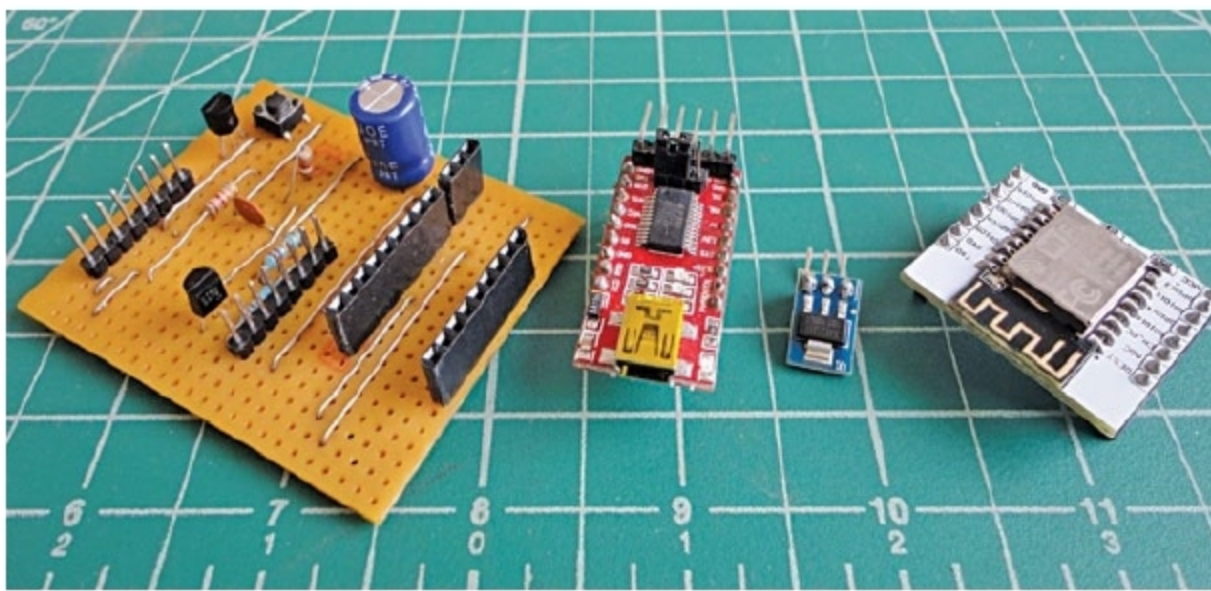


Fig. 7: Modules and general-purpose PCB used by author

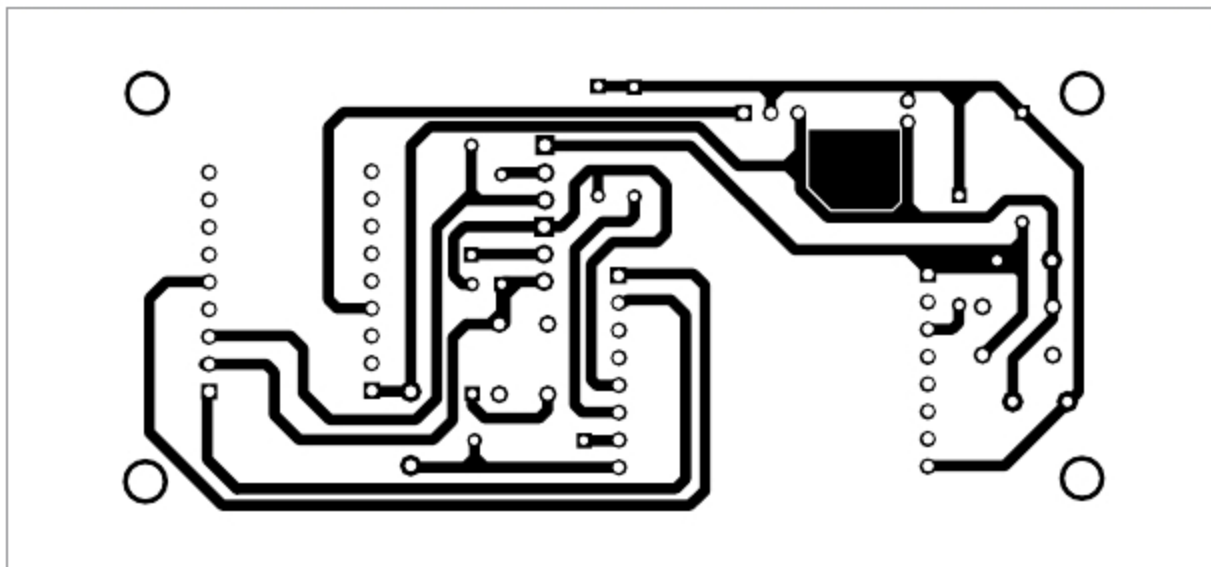


Fig. 8: Actual-size PCB layout of the automatic programmer

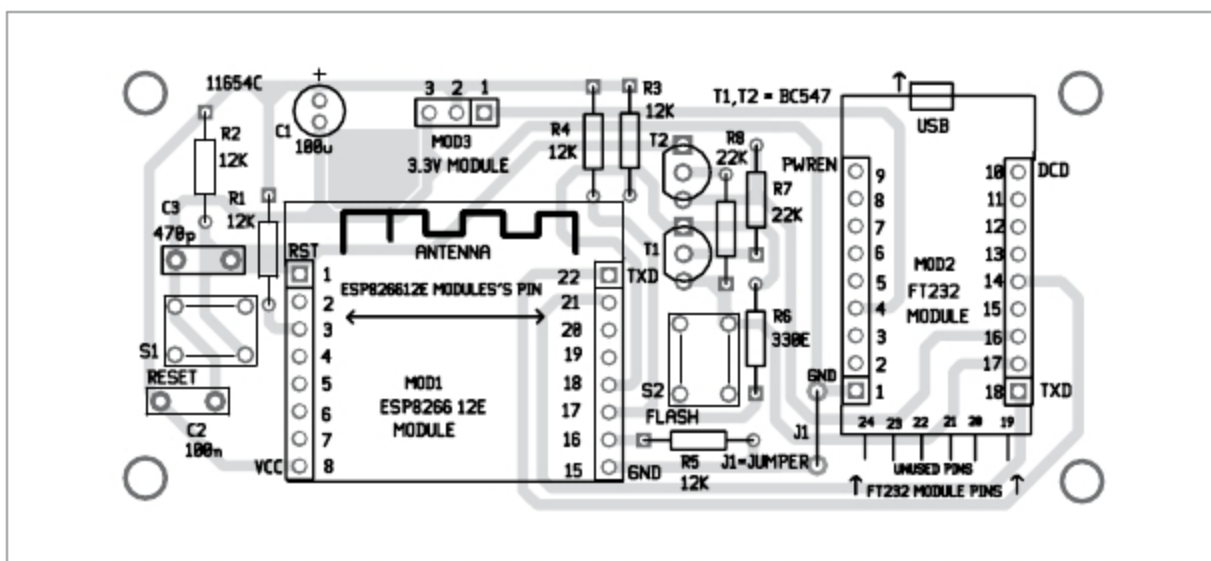


Fig. 9: Components layout for the PCB

mally be held high but connected to ground through micro switches, as shown in the circuit in Fig. 4. This is the minimum configuration necessary for using ESP-12E/F module. Every project built around this module should have these five 12k resistors. These resistors are prebuilt on NodeMCU board.

Manually flashing process. A manual ESP-12E/F programmer is shown in Fig. 4. For manually flash-

ing any sketch/code to ESP8266 chip, first put it in flash/programming mode step-wise as follows:

1. Press and hold Reset (S1) and Flash (S2) buttons.
2. Release S1 while holding S2 down. This is equivalent to power-on while GPIO0 is held low.
3. Release S2 after ESP8266 boots up.
4. Upload the software via Tx and Rx pins. Since computers no longer

have hardware serial ports, a USB-to-UART converter may be used. This conversion happens on NodeMCU via built-in CH340G or CP2102 USB-to-UART chip. FTDI USB-to-UART module, which is available easily, is used for this programmer.

5. Once the flashing process is completed, restart ESP8266 by pressing switch S1 to boot ESP8266 in normal mode.

Automating flashing process. While the above process requires manual intervention every time new software is flashed, we need the convenience of loading the software code similar to the experience of using NodeMCU. This process can be automated by using data terminal ready (DTR) and ready-to-send (RTS) signals of the USB-to-UART chip, which are used for asserting Reset and Flash (GPIO0) pins on ESP8266 chip.

DTR and RTS pins are conveniently accessible on the popular FTDI module built around FT232x USB-to-UART module (shown in Fig. 5). External USB-to-UART modules using CH340 chips are also available, but DTR and RTS pins are not accessible on these modules.

RTS signal on FT232x USB-to-UART module is available on a separate pin, as shown in Fig. 5. For ease of use with the programmer, mount male header pins to the module where all the required outputs (holes) are available. An optional serial adaptor, consisting of six pins, is mounted horizontally on the module, rather than vertically.

FT232 module also has an integrated 3.3V level converter for USB I/O. Since ESP8266-12x works on 3.3V, jumper on the module should be set to 3.3V, so that signals are compatible with ESP8266. 3.3V can also be used to supply external logic. However, maximum current rating in FT232 is 100mA, while ESP8266-12E/F can draw peak currents of 170mA. To prevent damage to FT232 device, an external 3.3V module is used to power ESP8266-12E/F.

The programmer circuit

The circuit diagram of the automatic programmer is shown in Fig. 6. It is built around ESP8266-12E module